Sahaj Bhadja

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Summary

Software Engineer with 2+ years building and debugging **Java/Python backend systems**, writing **automated tests**, integrating services, and improving release quality through structured SDLC practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Designed and optimized **SQL-backed services** for 500+ users, reducing dashboard latency from 1.2s to 350ms by refining schemas, queries, and database access patterns.
- Built release validation workflows for 5K+ monthly jobs with reconciliation checks and defect logging, improving production stability, support handoffs, and confidence before business-facing deployments.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed and modified **Flask + React** features serving 10K+ daily requests, translating user requirements into modular APIs, maintainable components, and testable workflow boundaries.
- Implemented concurrent processing controls and CI/CD checks via GitHub Actions, reducing turnaround time by 60% while preserving API reliability and predictable behavior during peak traffic.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered API and ETL integrations using OOP, AWS Glue, and S3, improving response times by 30% and enabling cleaner integration into updated system environments.
- Created and maintained 20+ unit and integration suites with Jenkins, PyTest, and Mockito across three OS environments, improving release quality and reducing regression escapes.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed REST API and WebSocket modules supporting eight operational workflows and 2M+ synchronized records, adding structured logging to simplify debugging and defect reproduction.
- Automated build, test, and deployment flows with Jenkins and Docker, cutting deployment time by 50% and improving consistency across iterative releases and environment changes.

LST Capitals **India**
Web Developer Intern *November 2022 - January 2023*
- Built backend API integrations and data workflows processing 50K+ records, enabling faster feature iteration and dependable data exchange across payments and booking subsystems.
- Implemented webhook-driven automation and operational checks, improving issue traceability, documentation quality, and cross-team troubleshooting for production scenarios requiring rapid corrective fixes.

Skills

Languages: **Java**, **Python**, SQL, JavaScript/TypeScript, C++
Backend & APIs: Flask, FastAPI, Node.js, Express.js, REST APIs, JSON, OOP
QA & Automation: Unit testing, integration testing, CI/CD, Jenkins, PyTest, Mockito, defect triage
Platforms: Linux workflow, shell scripting, Git, GitHub Actions, Docker, Kubernetes
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL
Engineering Practices: SDLC, Agile/Scrum, debugging, technical documentation, release support

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built a **Java**-based SQL workspace spanning 10+ databases with schema-aware query generation, execution validation, and structured outputs that accelerated analysis and reduced manual query errors.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed API-integrated healthcare workflows supporting 100+ daily appointments, adding monitoring-backed operational checks that improved reliability, accelerated issue diagnosis, and reduced record retrieval latency by 2x.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented six scheduling algorithms with interactive controls and visual diagnostics across 1K+ simulated process combinations, strengthening algorithmic reasoning, debugging practices, and system-performance interpretation skills.